As the waves flow outward, the hole's deformation gradually grows weaker and the waves weaken.

When these waves reach Earth, they have the form that I show in the upper part of Figure 16.8. They stretch along one direction and squeeze along the other. The stretch and squeeze oscillate (from red right-left to blue right-left to red right-left, etc.) as the waves pass through the detector in the bottom part of Figure 16.8.

The detector consists of four huge mirrors (40 kilograms, 34 centimeters in diameter) that hang from overhead supports at the ends of two perpendicular arms. The waves' tendex lines stretch one arm while squeezing the other, and then squeeze the first while stretching the second, over and over and over again. The oscillating separation between mirrors is monitored with laser beams, by a technique called interferometry. Hence LIGO's name: Laser Interferometer Gravitational Wave Observatory.